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LOKNETE MA. HANMANTRAO PATIL CHARITABLE TRUST'S
ADARSH INSTITUTE OF TECHNOLOGY
& RESEARCH CENTRE
ENGINEERING & POLYTECHNIC
AN AUTONOMOUS INSTITUTE
Approved by AICTE New Delhi & DTE Mumbai, Affiliated to Dr. Babasaheb Ambedkar Technological University, Lonere & MSBTE, Mumbai

Department of Artificial Intelligence and Machine Learning

COURSE STRUCTURE & CREDIT DISTRIBUTION

(To Be Implemented from AY 2025-26 Batch)

ABOUT DEPARTMENT
Artificial Intelligence and Machine Learning Engineering is a key department of the Adarsh Institute of Technology and Research Centre, Vita, established in 2022. The department has highly qualified faculty and well-equipped laboratories with modern instruments to meet the growing needs of the industry. With the increasing demand for skilled engineers in Government and private sectors, the department is committed to producing competent, innovative, and industry-ready professionals. Equal emphasis is placed on academic excellence and practical exposure, with students gaining valuable experience through industrial training, internships, and live projects to prepare them for real-world challenges. Technical workshops, expert lectures, and research activities are regularly conducted to enhance knowledge and skills. For extracurricular and technical engagement, the student association Artificial Intelligence and Machine Learning Student Association (AMESA), organizes various technical, cultural, and social events throughout the year, fostering leadership, teamwork, and innovation. “Artificial Intelligence and Machine Learning Engineers shape the future with knowledge and creativity.”

DEPARTMENT VISION
To excel in the artificial intelligence & machine learning discipline through continuous research and industry oriented curriculum leading to responsible AI professionals.

DEPARTMENT MISION	
M1	To inculcate teaching and learning processes in artificial intelligence & machine learning to address global challenges.
M2	To integrate research and entrepreneurship skills to address present and future challenges of the society and industry.
M3	To establish state of art laboratories and centre of excellence in the emerging fields of AI and ML.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)	
PEO1	To develop expertise in AI & ML field which will deliver technically feasible, commercially viable and socially acceptable solutions to real-world problems.
PEO2	Graduates will perform their duties with professional skills and ethics.
PEO3	Graduates will demonstrate lifelong learning in the field of AI & ML by up-gradation of knowledge, skills and abilities.

PROGRAM OUTCOMES (POs)	
PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development.
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8)
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).
PO7	**Ethics:** Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)
PO8	**Individual and Collaborative Team work:** Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO9	**Communication:** Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences
PO10	**Project Management and Finance:** Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.
PO11	**Life-Long Learning:** Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOMES (PSOs)	
PSO1	Apply knowledge of artificial intelligence & machine learning to identify, formulate and solve complex engineering problems.
PSO2	Demonstrate use of AI and ML techniques for solutions of real life problems.
PSO3	Design and develop applications of AI & ML for Data Analytics, Health care systems and allied domains.

Department of Artificial Intelligence and Machine Learning

A. Definition of Credit:

Details	Credits
1 Hr. Lecture (L) per week	1 Credit
1 Hr. Tutorial (T) per week	1 Credit
1 Hr. Practical (P) per week	0.5 Credit
2 Hours Practical (P) per week	1 Credit

B. Range of Credits:

A range of credits from 160 to 176 for a student to be eligible to get undergraduate degree in engineering. Student will be eligible to get undergraduate degree with minors 14 credits, if one completes additional credits.

C. Structure & Definition:

Structure Code and Definition		
Sr. No	**DEFINITION**	**CODE**
1	Basic Science Course	BSC
2	Engineering Science Course	ESC
3	Program Core Courses	PCC
4	Program Elective Course	PEC
5	Multidisciplinary Minor	MDM
6	Open Elective	OE
7	Vocational and Skill Enhancement Course	VSEC
8	Humanities social sciences and Management	HSSM
9	Experiential Learning courses	ELC
10	Co-curricular Courses	CC

D. Credit Framework:

Level	Qualification Title	Credit Requirements (Minimum)	Credit Requirements (Maximum)	Semester	Year
6.0	4-Years Bachelor's degree (B.E./ B.Tech. or Equivalent) in Engg./ Tech. with Multidisciplinary Minor	160	176	8	4
6.0	4-Years Bachelor's degree (B.E./ B.Tech. or Equivalent) in Engg./ Tech. – Honors and Multidisciplinary Minor	180	194	8	4

E. Structure Categories of UG Program:

Sr. No	Year	FY		SY		TY		B.Tech		Actual	NEP Guidelines	Actual %
	Semester	I	II	III	IV	V	VI	VII	VIII			
1	BSC	7	7	3	3					20	14-16	11.36
2	ESC	7	7							14	12-16	7.95
3	PCC			17	18	9	8	5		57	44-56	32.38
4	PEC					3	3	3	4	13	20	7.38
5	OE					3	3	3		9	8	5.11
6	MDM			3	3	3	3	3		15	14	8.52
7	VSEC	3	4	2	1	1				11	8	6.25
8	HSSM	3	2			4	3			12	14	6.81

9	CC	1	1							2	4	1.14
10	ELC					1	3	7	12	23	22	13.06
Total Credits		**21**	**21**	**25**	**25**	**24**	**23**	**21**	**16**	**176**	**160-176**	**100**
Examination Total		**950**	**950**	**900**	**950**	**950**	**1000**	**850**	**400**	**6950**		
Total Working Hrs/Week		**26**	**26**	**30**	**30**	**28**	**27**	**26**	**28**	**221**		
NEP GUIDELINES		**20-22**	**20-22**	**20-22**	**20-22**	**20-22**	**20-22**	**20-22**	**20-22**	**160-176**		

Department of Artificial Intelligence and Machine Learning

Second Year B. Tech (S.Y.)

Semester: III

Sr. No.	Course Category	Course Code	Course Title	Teaching Scheme			Evaluation Scheme (% marks)				Cred-its
				L	T	P	CA	MSE	ESE	Total	
1	**BSC**	AM0301	Engineering Mathematics-III	3	0	0	20	20	60	100	3
2	**PCC**	AM0302	Discrete Mathematics	3	0	0	20	20	60	100	3
3	**PCC**	AM0303	Data Structures	3	0	0	20	20	60	100	3
4	**PCC**	AM0304	Fundamental of AI	3	0	0	20	20	60	100	3
5	**PCC**	AM0305	Operating System	3	0	0	20	20	60	100	3
6	**MDM**	AM0306	Multi-Disciplinary Minor-01	3	0	0	20	20	60	100	3
7	**VSEC**	AM0307	Web Development Laboratory	1	0	2	50	-	-	50	2
8	**PCC**	AM0308	Data Structures Laboratory	0	0	2	50	-	50	100	1
9	**PCC**	AM0309	Python Programming Laboratory	1*	0	4	50	-	50	100	3
10	**PCC**	AM0310	Fundamental of AI Laboratory	0	0	2	50	-	-	50	1
Total				**20**	**0**	**10**	**320**	**120**	**460**	**900**	**25**

Note: * indicates Integrated Course

Second Year B. Tech (S.Y.)

Semester: IV

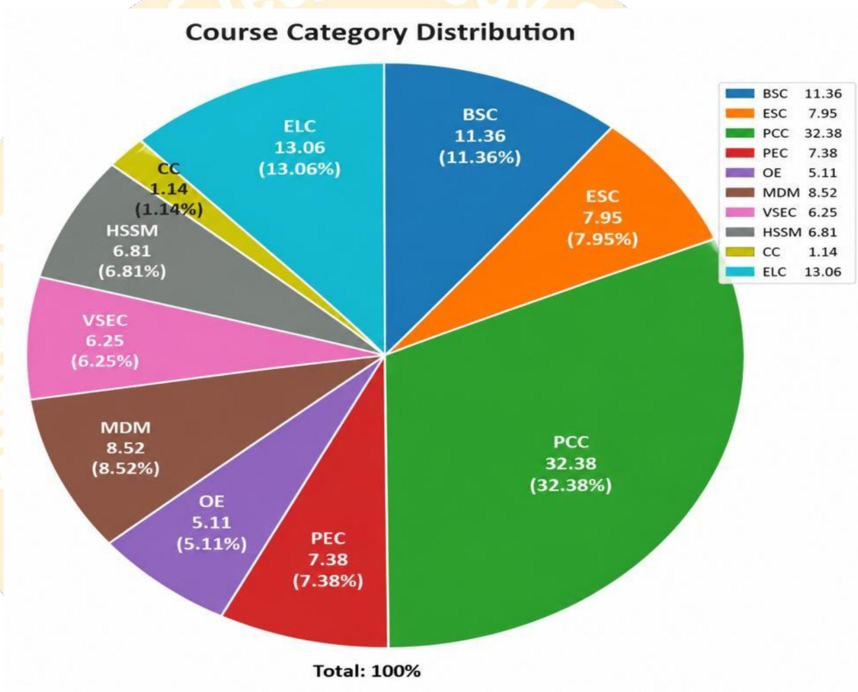
Sr. No.	Course Category	Course Code	Course Title	Teaching Scheme			Evaluation Scheme (% marks)				Cred-its
				L	T	P	CA	MSE	ESE	Total	
1	**BSC**	AM0401	Probability and Statistics	3	0	0	20	20	60	100	3
2	**PCC**	AM0402	Database Management System	3	0	0	20	20	60	100	3
3	**PCC**	AM0403	Design and Analysis of Algorithm	3	0	0	20	20	60	100	3
4	**PCC**	AM0404	Cloud Computing	3	1	0	20	20	60	100	4
5	**MDM**	AM0405	Multi-Disciplinary Minor-02	3	0	0	20	20	60	100	3
6	**PCC**	AM0406	Machine Learning	3	0	0	20	20	60	100	3
7	**PCC**	AM0407	Machine Learning Laboratory	0	0	2	50	-	50	100	1
8	**PCC**	AM0408	Java Programming Laboratory	1*	0	4	50	-	50	100	3
9	**PCC**	AM0409	Database Management System Laboratory	0	0	2	50	-	50	100	1
10	**VSEC**	AM0410	Mobile Application Development Laboratory	0	0	2	50	-		50	1
Total				**19**	**1**	**10**	**320**	**120**	**510**	**950**	**25**

Note: * indicates Integrated Course

Exit Course

Exit option: Award of UG Diploma in Major with 92 credits and an additional 8 credits from following Exit Courses				
Sr. No	**Course Code**	**Course Title**	**Mode**	**Credits**
1	AM-EC-0201	Generative AI:Prompt Engineering Basics	Online/offline certification Course	8
		OR		
2	AM-EC-0202	Fundamental of Machine Learning and Artificial Intelligence	Online/offline certification Course	8

G. Category wise Credit Distribution



H. Multi-Disciplinary Minor Courses Offered:

MDM (Bucket) offered by Artificial Intelligence & Machine Learning

Sr. No	MDM Bucket	Course Code	Name of Course
1	Multi-Disciplinary Minor-01	AM0306	Data Analytics Using Python
2	Multi-Disciplinary Minor-02	AM0405	Machine Learning
3	Multi-Disciplinary Minor-03	AM0505	Advance Machine Learning
4	Multi-Disciplinary Minor-04	AM0605	Deep Learning
5	Multi-Disciplinary Minor-05	AM0705	MLOps

MDM (Bucket) offered by Civil Engineering

Sr. No	MDM Bucket	Course Code	Name of Course
1	Multi-Disciplinary Minor-01	CE0306	Fundamentals of Civil Engineering
2	Multi-Disciplinary Minor-02	CE0406	Building Materials & Construction
3	Multi-Disciplinary Minor-03	CE0505	Building Estimates & Costing
4	Multi-Disciplinary Minor-04	CE0606	Repair & Maintenance
5	Multi-Disciplinary Minor-05	CE0705	GIS and Remote Sensing

MDM (Bucket) offered by Computer Science Engineering

Sr. No	MDM Bucket	Course Code	Name of Course
1	Multi-Disciplinary Minor-01	CS0306	Introduction to Python Programming
2	Multi-Disciplinary Minor-02	CS0406	Object Oriented Programming using Java
3	Multi-Disciplinary Minor-03	CS0505	Database Systems
4	Multi-Disciplinary Minor-04	CS0605	Operating Systems
5	Multi-Disciplinary Minor-05	CS0705	Introduction to Data Science

MDM (Bucket) offered by Electrical Engineering

Sr. No	MDM Bucket	Course Code	Name of Course
1	Multi-Disciplinary Minor-01	EE0305	Electrical and Electronics Measurements
2	Multi-Disciplinary Minor-02	EE0405	Electrical Machines
3	Multi-Disciplinary Minor-03	EE0506	Power System
4	Multi-Disciplinary Minor-04	EE0606	Electrical Vehicle
5	Multi-Disciplinary Minor-05	EE0705	Industrial Automation & Control

MDM (Bucket) offered by Electronics & Telecommunication Engineering

Sr. No	MDM Bucket	Course Code	Name of Course
1	Multi-Disciplinary Minor-01	ET0306	Digital Electronics
2	Multi-Disciplinary Minor-02	ET0406	Signals and Systems
3	Multi-Disciplinary Minor-03	ET0505	Microcontroller and Application
4	Multi-Disciplinary Minor-04	ET0605	Digital Communication
5	Multi-Disciplinary Minor-05	ET0704	Mobile Communication

MDM (Bucket) offered by Mechanical Engineering

Sr. No	MDM Bucket	Course Code	Subject Name
1	Multi-Disciplinary Minor-01	ME0305	Smart Materials
2	Multi-Disciplinary Minor-02	ME0405	Product Design and Prototyping
3	Multi-Disciplinary Minor-03	ME0505	Additive Manufacturing
4	Multi-Disciplinary Minor-04	ME0605	Industrial Robotics and Automation
5	Multi-Disciplinary Minor-05	ME0705	Enterprise Recourse Planning

Engineering Mathematics-III

AM0301	BSC	3L - 0T - 0P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 3hrs/week Tutorial: 1 Hrs./week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites:
Basics of Integral transforms and Vector Calculus knowledge

Course Description : This course equips engineering students with advanced mathematical tools to model, analyze, and solve complex technological problems. Converts differential equations into simpler algebraic equations, Deconstructs complex waveforms into sinusoidal components, vital for signal processing, Extends calculus to vector fields to analyze fluid flow, electromagnetism, and mechanics using gradient, divergence and curl, Utilizes mathematical models and algorithms (like the Simplex method).

Course Objectives:	
1	Able to comprehend the fundamental knowledge of the Laplace and inverse Laplace transforms and their derivatives for elementary functions.
2	Able to apply the properties of Laplace and inverse Laplace transforms to solve simultaneous and linear differential equations with constant coefficients
3	Recognize the importance of inverse Laplace transforms in control systems, signal processing, and other engineering applications.
4	Able to conceptualize the definitions and properties of Fourier transforms and to solve boundary value problems using Fourier transforms.
5	Understand vector differentiation concepts like gradient, divergence, and curl, and apply them to engineering problems.
6	Learn vector integration techniques including line, surface, and volume integrals, and apply them to solve engineering problems

Course Outcomes: On completion of the course, students will be able to,	
CO1	Use Laplace transforms to solve initial value problems in engineering systems.
CO2	Evaluate inverse Laplace transforms using various methods (partial fractions, convolution, etc.).
CO3	Solve differential equations and physical systems using inverse Laplace methods.
CO4	Understand and apply Fourier transform to analyze signals in frequency domain.
CO5	Apply vector differentiation concepts to problems in fluid mechanics, electromagnetics, and other physical fields.
CO6	Solve two variable LPP using graphical method and interpret feasible region

CO - PO Mapping														
Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	–	2	1	–	–	–	–	–	2	2	1	-
CO2	3	2	–	2	1	–	–	–	–	–	2	2	1	-
CO3	3	3	1	2	1	–	–	–	–	–	2	3	2	1
CO4	3	2	–	2	2	–	–	–	–	–	2	2	2	1
CO5	3	3	1	2	1	–	–	–	–	–	2	2	1	-
CO6	3	3	1	3	1	–	–	–	–	–	2	2	1	-

Unit No.	Course Contents	Hrs.
1	**Laplace Transform Definition** - conditions for existence, transforms of elementary functions, linearity property, first shifting property, second shifting property, , transforms of functions multiplied by tn , scale change property, transforms of function divided by t, transforms of integral of functions, transforms of derivatives, evaluation of integrals by using Laplace transform; Transforms of some special functions- transforms of periodic function, Heaviside-unit step function, Dirac delta function.	6
2	**Inverse Laplace Transform** Introductory remarks, inverse transforms of some elementary functions, general methods of finding inverse transforms, partial fraction method.	6
3	**Methods of Finding Inverse Laplace** Convolution Theorem for finding inverse Laplace transforms, applications to find the solutions of simultaneous linear differential equations with constant coefficients.	6
4	**Fourier Transform** Definitions - integral transforms, Fourier integral theorem (without proof), Fourier sine and cosine integrals, complex form of Fourier integrals, Fourier sine and cosine transforms, properties of Fourier transforms, Parseval’s identity for Fourier Transforms.	6
5	**Vector Differentiation** Differentiation of vectors, Gradient of scalar point function, Directional derivative, Divergence of vector point function, Curl of a vector point function, Irrotational and solenoidal vector field.	6
6	**Optimization &Linear Programming** Introduction: Concept of optimization, formulation of optimization problems. Linear programming problem(LPP): formulation, graphical method of two variables. Feasible solution, basic feasible solution, optimal solution	6

References-

Text Books:

1. Higher Engineering Mathematics,B. S. Grewal,Khanna Publishers.
2. Higher Engineering Mathematics, H. K. Das and Er. Rajnish Verma,S. Chand. & CO. Pvt. Ltd.

Reference Books:

1. Applied Mathematics. V. Kumbhojkar,C. Jamnadas & Co.Vol. III.
2. Integral Transforms and their Engineering Applications,Dr. B. B. Singh Synergy Knowledgeware.

Discrete Mathematics

AM0302	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	**Examination Scheme**
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Probability Theory, Set Theory.

Course Description: This course introduces the fundamental concepts of discrete mathematical structures used in computer science and engineering. It covers topics such as logic, sets, relations, functions, combinatorics, graph theory, and recurrence relations. Students will develop analytical and problem-solving skills through mathematical reasoning and proofs.

Course Objectives:	
1	To apply basic concepts of set theory, logic, proof techniques, graphs and trees.
2	To analyze the basic concepts of relations and functions.
3	To learn the concepts of algebraic system & groups.

Course Outcomes: On completion of the course, students should be able to,	
CO1	Apply the logical reasoning to solve the problems.
CO2	Solve the problems based on binary relations
CO3	Describe the different terminologies in algebraic structures
CO4	Evaluate the problems based on Boolean functions.
CO5	Solve different graph problems like PERT graph, tree traversal.
CO6	Solve different problems on counting principle, permutation & combinations.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2										3	2	1
CO2	3	2										3	2	2
CO3	2											2	1	1
CO4	3	2										3	3	2
CO5	3	2										3	3	3
CO6	3	2										3	2	2

Unit No.	Course Contents	Hrs.
1	**Mathematical Logic** Statements and Notations, Connectives negation, Conjunction, disjunction, conditional, bi-conditional, Statement formulas and truth tables, well-formed formulas, Tautologies, Equivalence of formulas, Duality law, Tautological implications, functionally complete sets of connectives, other connective, Normal and principal normal forms,	7
2	**Set Theory** Basic concepts of set theory, Operations on sets, ordered pairs, Cartesian Products, Relation and Ordering properties of binary relations in a set, Relation matrix and the graph of a relation, Partition and Covering of set, Equivalence relations, Composition of Binary relations, Functions Cypes, composition of functions, Inverse functions. Practice Problems.	7
3	**Algebraic Systems** Algebraic systems, properties and examples, Semigroups and Monoids, properties and examples, Homomorphism of Semigroups and Monoids, Groups: Definition and examples, Subgroups and homomorphism, Practice Problems.	5
4	**Lattices and Boolean Algebra** Partial ordering, POSET and Hasse diagram, Lattice as POSETs, definition, examples and properties, Lattice as algebraic systems, Special lattices. Boolean algebra definition and examples, Boolean functions, representation and minimization of Boolean functions, Practice Problems.	6
5	**Graph Theory** Basic concepts of graph theory, Complete, Regular and Bipartite Graphs, Graph Coloring, Storage representation and manipulation of Graphs, PERT and related techniques.	5
6	**Permutations, Combinations and Probability theory** Random Experiments, Sample space & Events, Pigeon hole principle, Permutations and Combinations, Concept of Probability Discrete Probability, Conditional Probability**.**	6

References:

Text Book:

1. Discrete Mathematical Structures with Application to Computer Science by J. P. Tremblay & R. Manohar, MGH International
2. Elements of Discrete Mathematics by C. L. Líu and D. P. Mohapatra, Tata McGraw-Hill

Reference Book:

1. Discrete Mathematics by Seymour Lipschutz, Marc Lipson, MGH Schaum's outlines
2. Discrete Mathematics and its Applications by Kenneth H. Rosen, AT&T Bell Labs

Data Structures

AM0303	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic knowledge of C programming, Knowledge of basic mathematical concepts.

Couse Description:

This course introduces the fundamental concepts of data structures and algorithms for efficient data organization and processing. It covers stacks, queues, linked lists, searching, sorting, hashing, trees, and graphs. Students will learn various data representation techniques and operations to solve computational problems effectively.

Course Objectives:

1	To understand the fundamental concepts of data structures, Abstract Data Types (ADTs), and algorithms.
2	To learn the implementation and applications of stack and queue data structures using different techniques.
3	To understand linked list data structures and perform various operations on singly, doubly, and circular linked lists.
4	To study searching, sorting, and hashing techniques for efficient data organization and retrieval.
5	To learn tree data structures, binary tree traversal methods, and Binary Search Tree operations.
6	To understand graph representations and graph traversal algorithms such as BFS and DFS.

Course Outcomes: On completion of the course, students will be able to,

CO1	**Apply** basic data structure concepts, ADTs, algorithms, and operations.
CO2	**Apply** stack and queue ADTs using static, dynamic, circular, and priority implementations.
CO3	**Apply** singly, doubly, and circular linked lists to perform basic operations.
CO4	**Analyze** various searching and sorting algorithms and hashing techniques, including hash functions and collision resolution methods.
CO5	**Implement** tree data structures including binary trees, their representations, traversal techniques, and operations in Binary Search Trees such as insertion and deletion.
CO6	**Analyze** graph representations and traversal techniques such as adjacency matrix, adjacency list, BFS, and DFS.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	0	1	0	0	0	0	0	1	1	0	0
CO2	3	2	2	1	2	0	0	0	0	0	1	2	1	0
CO3	3	2	2	2	2	0	0	0	0	0	1	2	1	0
CO4	3	3	2	2	2	0	0	0	0	0	1	2	2	1
CO5	3	2	3	2	3	0	0	0	0	0	1	3	2	2
CO6	3	3	2	2	2	0	0	0	0	0	2	3	2	1

Unit No.	Course Contents	Hrs.
1	**Introduction to Data Structures:** Primitive and non-primitive data structures, Operations on data structures, Algorithms, Abstract Data Types (ADT).	5
2	**Stacks and Queues:** Stacks as ADT, operations, representation using static and dynamic structures, applications of stack Queue as ADT, operations, representation using static and dynamic structures, circular queue, priority queue, double ended queue.	7
3	**Linked List:** Definition and structure of singly linked list, doubly linked list and circular linked list. Operations: creation, traversal, insertion, deletion.	7
4	**Searching and Sorting Techniques:** Linear search, binary search, Internal and External Sorts, bubble sort, selection sort, insertion sort, merge sort, quick sort, radix sort, heap sort. Hashing - Definition, hash functions, overflow, collision, Collision resolution techniques, Open addressing, Chaining.	7
5	**Trees:** Definitions and concepts, Binary trees, Binary Tree Representations, Binary Tree Traversals, Binary Search Tree, Insertion and Deletion in BST	5
6	**Graph:** Basic concept of graph theory, storage representation: adjacency matrix, adjacency list, adjacency multi-lists, graph traversal techniques- BFS and DFS	5

References:

Text Book:

1. Mark Allen Weiss, Data Structures and Algorithm Analysis in C++, Pearson Education, 4th Edition, 2013, ISBN: 978-0132847377
2. Richard F. Gilberg, Behrouz A. Forouzan, Data Structures: A Pseudocode Approach with C, Cengage Learning, 2nd Edition, 2005, ISBN: 978-0534390808
3. Ellis Horowitz, Sartaj Sahni, Susan Anderson-Freed, Fundamentals of Data Structures in C, Universities Press, 2nd Edition, ISBN: 978-8173716058

Reference Book:

1. Seymour Lipschutz, Data Structures, McGraw-Hill Publication, Revised 1st Edition, 2014, ISBN: 978-0070257399
2. Y. Langsam, M. Augenstein, A. Tanenbaum, Data Structures Using C and C++, Prentice Hall India Learning Private Limited, 2nd Edition, 1998, ISBN: 978-8120311770
3. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, PHI Publication, 2nd Edition, 2002, ISBN: 978-0262032933

Fundamental of AI

AM0304	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic of computing and fundamental knowledge of problem solving techniques.

Course Description: This course provides an introduction to the principles, concepts, and applications of Artificial Intelligence (AI). Students will learn about intelligent agents, problem-solving techniques, knowledge representation, search algorithms, and machine learning fundamentals. The course explores how AI systems perceive, reason, learn, and make decisions in real-world environments.

Course Objectives:	
1	To understand the fundamentals, evolution, current state, and core concepts of Artificial Intelligence, including intelligent agents, rational behavior, environments, and agent structures.
2	To apply problem-solving techniques, including uninformed, informed, and adversarial search strategies, to develop optimal solutions for well-defined problems.
3	To represent and reason with knowledge using predicate logic, rules, and inference methods, and apply these to solve reasoning problems
4	To implement probabilistic reasoning, planning techniques, and natural language processing methods to handle uncertainty and language-based tasks.
5	To evaluate Artificial Intelligence-Ethics relationship.
6	To investigate future trends in Artificial Intelligence.

Course Outcomes: On completion of the course, students should be able to,	
CO1	Demonstrate Artificial Intelligence concepts and architecture for learning agents.
CO2	Apply problem-solving methods and search algorithms
CO3	Analyse the fundamentals of knowledge representation and reasoning.
CO4	Apply probabilistic reasoning and planning techniques to support intelligent decision-making in uncertain environments
CO5	Evaluate ethical implications of Artificial Intelligence technologies for Bias,Fairness,Privacy concern, Accountability and transparency
CO6	Investigate future trends in AI and create learners who will look for prospective applications

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	-	-	1	-	-	-	-	-	1	2	2	1
CO2	3	3	2	1	2	-	-	-	-	-	1	3	2	2
CO3	3	3	1	2	1	-	-	-	-	-	1	2	2	2
CO4	3	3	2	2	2	-	-	-	-	-	1	3	3	3
CO5	1	2	-	-	-	3	3	1	1	-	2	1	2	2
CO6	2	2	2	1	2	1	-	1	1	1	3	2	3	3

Unit No.	Course Contents	Hrs.
1	**Introduction:** What Is AI? The Foundations of Artificial Intelligence, the History of artificial Intelligence, the State of the Art. Intelligent Agents: Agents and Environments Good Behavior: The Concept of Rationality, The Nature of Environments, The Structure of Agents.	5
2	**Problem-solving and Searching:** Solving Problems by Searching, Problem-Solving Agents, Example Problems, Searching for Solutions, Uninformed Search Strategies,Informed (Heuristic) Search Strategies, Heuristic Functions, Defining Constraint Satisfaction Problems, Constraint Propagation: Inference in CSPs, Backtracking Search for CSPs, Local Search for CSPs, The Structure of Problems. Adversarial Search, Games, Optimal Decisions in Games, Alpha–Beta Pruning.	7
3	**Knowledge & Reasoning:** Knowledge representation issues, Representation & mapping, Approaches to knowledge representation, Issues in knowledge representation. Using predicate logic: Representing simple facts in logic, Representing instant & ISA relationship, Computable functions & predicates, Resolution, Natural deduction. Representing knowledge using rules: Procedural verses declarative knowledge, Logic programming, Forward verses backward reasoning, Matching, Control knowledge.	7
4	**Probabilistic Reasoning:** Representing knowledge in an uncertain domain, The semantics of Bayesian networks, Dempster-Shafer theory, Fuzzy sets & fuzzy logics, Planning: Overview, Components of a planning system, Goal stack planning, Hierarchical planning and other planning techniques.	5
5	**Ethics and Social Implications in AI** Bias and Fairness in AI Systems ,Privacy Concern, Accountability and Transparency in AI Ethical Implications of AI in Society **Case Studies:** 1. Tesla's Autopilot Incidents: Addressing safety, ethical, and regulatory Concerns arising from Autopilot incidents to improve trust, ensure user safety 2. Cambridge Analytica and Facebook :lessons can be learned from the Cambridge Analytica and Facebook data scandal about balancing user privacy, data ethics, and corporate accountability in the age of big data and social media platforms	6
6	**Future of Artificial Intelligence** Emerging Technologies in AI, Challenges in Achieving Artificial General Intelligence (AGI) Prospective Applications and Advancements **Case Studies:** 1. AI in Predicting Climate Change Impacts Investigate how AI-driven climate models are helping predicts environmental changes and their societal implications. 2. Smart Cities Powered by AI Examine AI-driven smart city projects, like Singapore's Smart Nation initiative, focusing on traffic management, urban planning, and sustainable living	6

References:

Text Book:

1. Artificial Intelligence: Concepts and Applications by Lavika Goel Latest Edition, Wiley.
2. Artificial Intelligence by Kevin Knight, Elaine Rich, Shiva Shankar B. Nair, Latest Edition, Mc Graw Hill.

Reference Books:

1. The Handbook of Artificial Intelligence by Avron Barr and Edward A, Heuris Tech Press
2. Fundamentals of the New Artificial Intelligence: Neural, Evolutionary Fuzzy and More by Toshinori Munakata, Springer

Operating System

AM0305	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic knowledge of computer organization, data structures, and C programming.

Couse Description:

This course provides an introduction to Operating System concepts, including process management, memory management, synchronization, deadlocks, file systems, and I/O management. It also covers scheduling techniques, system calls, and storage management. Additionally, the course introduces Agenting Operating Systems and their applications in intelligent computing environments.

Course Objectives:

1	To understand the basic concepts, structures, and functions of operating systems.
2	To learn process management, process synchronization, and CPU scheduling techniques.
3	To understand deadlock handling methods and memory management strategies used in operating systems.
4	To study file systems, storage management, and I/O subsystem operations.
5	To understand multithreading, interprocess communication, and kernel functionalities.
6	To learn the concepts, architecture, applications, advantages, and limitations of Agenting Operating Systems and intelligent agents.

Course Outcomes: On completion of the course, students will be able to,

CO1	**Explain** the fundamental concepts of operating systems, system structures, services, and types of kernels.
CO2	**Analyze** process management concepts including process scheduling, multithreading, and interprocess communication.
CO3	**Apply** synchronization techniques to solve critical section problems and classical synchronization issues.
CO4	**Demonstrate** memory management techniques including paging, segmentation, and contiguous memory allocation.
CO5	**Describe** file systems, storage management, and I/O subsystem concepts including file access, protection, and kernel I/O operations.
CO6	**Analyze** the architecture, communication, security, applications, advantages, and limitations of Agenting Operating Systems.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	0	0	1	0	0	0	0	0	1	0	0	0
CO2	3	3	2	1	2	0	0	0	0	0	1	2	1	0
CO3	3	3	2	2	2	0	0	0	0	0	1	1	1	0
CO4	3	3	2	2	2	0	0	0	0	0	1	1	0	0
CO5	3	2	2	1	2	0	0	0	0	0	1	1	1	1
CO6	3	2	2	1	3	0	0	0	0	0	1	1	2	2

Unit No.	Course Contents	Hrs.
1	**Overview:** Introduction to operating Systems, system structures, what operating systems do, computer system organization, computer system architecture, Operating - System structure, operating System operations, Types of Operating Systems, Operating System Services, User- operating system interface, System calls, Types of system calls, system programs, Kernel, Types of kernel.	**7**
2	**Process management:** Process concept: Process scheduling, operations on processes, Interprocess communication, Multi-Threaded programming: overview, Multithreading models, Thread Libraries, Threading issues, process scheduling: Basic concepts, Scheduling criteria, Scheduling algorithms, Multiple-Processor scheduling, Thread Scheduling.	**7**
3	**Process Synchronization:** Background, the critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of Synchronization	**5**
4	**Deadlocks:** System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm. **Memory Management:** Memory management strategies: Background; Swapping; Contiguous memory allocation; Paging; Structure of page table; Segmentation.	**6**
5	**Storage Management and I/O Subsystem**: File System: File concept, access methods, directory and disk structure, file-system mounting, file sharing, protection, Overview of I/O system, I/O hardware, Application I/O interface, Kernel I/O subsystem.	**5**
6	**Introduction to Agenting OS:** Introduction to Intelligent Agents, Architecture of Agenting OS, Functions of Agenting OS, Agent Communication and Coordination, Security in Agenting OS, Applications of Agenting OS, Advantages of Agenting OS, Limitations of Agenting OS.	**6**

References:

Text Book:

1. Silberschatz, Galvin, Operating System Concept, John Wiley, 8th Edition, 2009, ISBN: 978-0470128725
2. Dhananjay M. Dhamdhere, Operating Systems Concepts and Design, Tata McGraw Hill, 2nd Edition, 2006, ISBN: 978-0070616134
3. Dhananjay M. Dhamdhere, Operating Systems: A Concept-Based Approach, Tata McGraw Hill, 2nd Edition, 2007, ISBN: 978-0070668188
4. Ann McHoes & Ida M. Flynn, Understanding Operating Systems, Thomson, 7th Edition, 2014, ISBN: 978-128544552

Reference Book:

1. Charles Crowley, Operating System: A Design-Oriented Approach, Tata McGraw Hill, 1st Edition, 2001, ISBN: 978-0072465631
2. Achyut S. Godbole, Operating System with Case Studies in Unix, Netware and Windows NT, Tata McGraw Hill, 2nd Edition, 2007, ISBN: 978-0070616134
3. William Stallings, Operating Systems: Internals and Design Principles, Pearson Education International, 8th Edition, 2014, ISBN: 978-0133805918

Data Analytics Using Python

AM0306	Multi-Disciplinary Minor-01	3L - 0T - 0P	3 Credits

Teaching Scheme	**Examination Scheme**
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic python programming Language Knowledge

Course Description: This course introduces the fundamental concepts and techniques of data analytics using Python programming. Students will learn how to collect, clean, analyze, and visualize data using libraries such as NumPy, Pandas, Matplotlib, and Seaborn. The course covers exploratory data analysis, statistical methods, and data-driven decision-making.

Course Objectives:

1	To understand Python fundamentals, including syntax, data types, control structures, functions, OOP concepts, and sequence data structures for developing basic Python programs.
2	To understand the concepts, types, and life cycle of Data Analytics and apply core Python libraries such as NumPy, Pandas, Matplotlib, and Seaborn for analytical tasks.
3	To create and manipulate multidimensional NumPy arrays using indexing, slicing, reshaping, and transformation techniques for efficient numerical data processing.
4	To perform data loading, cleaning, transformation, and analysis using Pandas data structures such as Series and DataFrames.
5	To visualize data effectively using Matplotlib, Pandas plotting functions, and interactive visualization tools such as Bokeh and Plotly.
6	To apply data aggregation, grouping, pivot tables, and split-apply-combine techniques for summarizing and analyzing large datasets.

Course Outcomes: On completion of the course, students should be able to,

CO1	Apply Python fundamentals for data handling
CO2	Analyze data analytics concepts and processes
CO3	Use NumPy for numerical data computation
CO4	Perform data manipulation using Pandas
CO5	Visualize data using appropriate tools
CO6	Analyze datasets using aggregation techniques

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	–	–	3	–	–	–	–	–	2	3	3	2
CO2	2	3	–	2	2	–	–	–	–	–	2	3	2	2
CO3	2	3	2	2	3	–	–	–	–	–	1	3	3	2
CO4	2	3	3	2	3	–	–	–	–	–	1	3	3	3

CO5	1	2	3	–	3	–	–	–	2	–	1	2	3	3
CO6	2	3	2	3	3	–	–	–	–	–	1	3	3	3

Unit No.	Course Contents	Hrs.
1	**Getting started with Python:** Basics of Python including data types, operators, variables, expressions, control structures, conditional statements, looping structure, functions, OOP concepts. Python sequence data structures including String, Array, List, Tuple, Set, and Dictionary and associated operations	7
2	**Introduction to Data Analytics**. Data Analytics: An Overview, Importance of Data Analytics 1.2 Types of Data Analytics: Descriptive Analysis, Diagnostic Analysis, Predictive Analysis, Prescriptive Analysis, Visual Analytics, Life cycle of Data Analytics, Quality and Quantity of data, Measurement, Python Libraries: NumPy, Pandas, Matplotlib, Seaborn	6
3	**Array manipulation using Numpy:** Numpy array: Creating Numpy arrays; various data types of Numpy arrays, indexing and slicing, swapping axes, transposing arrays, data processing using Numpy arrays.	6
4	**Data Manipulation using Pandas:** Data Structures in Pandas: Series, DataFrame, Index objects, loading data into Panda's data frame, Working with Dataframes: Arithmetics, Statistics, Binning, Indexing, Filtering, handling missing data, Hierarchical indexing, Data wrangling: Data cleaning, transforming, merging and reshaping	7
5	**Plotting and Visualization:** Using Matplotlib to plot data: figures, subplots, markings, color and line styles, labels and legends, plotting functions in Pandas: Line, bar, Scatter plots, histograms, stacked bars, Heatmap, 3D Plotting, interactive plotting using Bokeh and Plotly.	5
6	**Data Aggregation and Group operations:** Group by mechanics, Data aggregation, General split-apply-combine, Pivot tables and cross tabulation	5

References:

Text Book:

1. Murach's Python Programming by Michael Urban and Joel Murach, Murach's Publication, 2016.
2. Core Python Programming by Dr. R. Nageswara Rao, Dreamtech Press, 1st Edition, 2016.
3. Python for Informatics: Exploring Information by Charles Severance, University of Michigan, 2014.

Reference Book:

1. Python Made Easy: Step by Step Guide to Programming and Data Analysis using Python for Beginners and Intermediate Level by Notion Press.,2020
2. Introduction to Computing & Problem Solving With PYTHON by Josef, J., Lal, S. P. Khanna book publishing Company (P) Limited.2016

Web Development Laboratory

AM0307	VSEC	1L - 0T - 2P	2 Credits

Teaching Scheme	Examination Scheme
Practical: 2hrs/week	CA: 50 Marks, MSE: - ESE: - Total: 50 Marks

Pre requisites:
Basic knowledge of computer systems and fundamental programming concepts.

Couse Description:
This course introduces modern web development using HTML, CSS, JavaScript, and React JS for building interactive and responsive web applications. It focuses on frontend development, DOM manipulation, API integration, state management, and deployment of real-world CRUD-based applications.

Course Objectives:	
1	To Introduce students to modern web development tools, HTML, CSS, and JavaScript for designing interactive web pages.
2	To Enable students to create structured, responsive, and user-friendly web interfaces using HTML5, CSS3, Flexbox, Grid, and Media Queries.
3	To Develop programming skills in JavaScript including functions, events, validation, and DOM manipulation for dynamic web applications.
4	To Familiarize students with React JS concepts such as components, props, state, hooks, lifecycle methods, and routing.
5	To Integrate APIs, manage application state using Redux, and develop responsive frontend applications.
6	To Deploy real-world CRUD-based web applications using React JS.

Course Outcomes: After Completion of the course students should able to,	
CO1	**Design** static and responsive web pages using HTML, CSS, Flexbox, Grid, and media queries.
CO2	**Implement** interactive client-side functionalities using JavaScript, events, validation, and DOM manipulation techniques.
CO3	**Develop** reusable and dynamic user interfaces using React JS components, props, state, hooks, and lifecycle methods.
CO4	**Implement** routing, API integration, asynchronous operations, and global state management in React applications.
CO5	**Create** responsive web applications with form handling, validation, and user interaction features.
CO6	**Deploy** CRUD-based real-world web applications such as student, employee, library, and contact management systems using React JS.

CO - PO Mapping														
Course Outcomes	**Program Outcomes**													
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	2	-	3	-	-	-	2	-	-	2	-	-
CO2	3	3	2	1	3	-	-	-	2	-	-	2	2	-
CO3	3	2	3	1	3	-	-	1	2	-	1	3	2	2
CO4	3	3	3	2	3	-	-	1	2	1	1	3	3	2
CO5	2	2	3	1	3	1	1	1	2	-	1	2	2	2
CO6	3	3	3	2	3	1	1	2	2	2	1	3	3	3

Pract No.	**List of Practical's**	**Hrs.**
1.	**Introduction to Web Development Tools** • Setup and Configuration of Visual Studio Code • Introduction to VS Code. **Basic HTML Page Design** • Create a static web page using HTML • Use headings, paragraphs, lists, links, and images	02
2.	**HTML Forms and Input Controls** • Design a registration form using HTML • Use input types, textarea, radio buttons, checkboxes	02
3.	**CSS Styling – Inline, Internal, External** • Apply CSS styles to HTML pages • Use colors, fonts, margins, padding, borders **Page Layout using CSS** • Design a webpage layout using div, flexbox, or grid • Create header, footer, navigation, and content sections	02
4.	**Title: JavaScript Basics** • Write JavaScript programs for variables, operators, conditions, loops **JavaScript Functions and Events** • Implement form validation using JavaScript • Handle events like onclick, onsubmit, onmouseover	02
5.	**DOM Manipulation** • Manipulate HTML elements using JavaScript DOM • Change content, style : innerHTML, innerText, textContent • Attributes dynamically : src (image), href (link), class, id, setAttribute()	02
6.	**Creating Student Profile Card using React** • Displaying student information in a college management system • Use JSX, Components, Props, CSS in React	02
7.	**Developing Counter Application using Class Component** • Visitor counter system in malls or events • Class Component, State, Event Handling	02

8.	**Implementing Lifecycle Methods in React** • Online/offline user status monitoring system • componentDidMount(), componentDidUpdate(), componentWillUnmount()	02
9.	**Creating Registration Form with Validation** • User signup form for websites • Form Handling, Validation, Controlled Components **Developing To-Do List using React Hooks** • Daily task management application • useState(), useEffect(), Functional Components	02
10.	**Implementing Routing in React** • Navigation system in e-commerce websites • React Router DOM, Routes, Navigation	02
11.	**Fetching Data from API in React** • Weather forecast application • API Integration, Axios/Fetch API, Async-Await	02
12.	**Deploy a simple CRUD (Create, Read, Update, Delete) application in React JS** that allows users to manage a list of items (for example, "Users"). The application should provide a user-friendly interface to add, view, edit, and remove users. **Student Management System** Add, view, update, and delete student records in a college system. **Employee Management System** Manage employee details such as name, department, salary, and designation. **Library Book Management** Add new books, update book details, view available books, and remove old books. **Contact Management Application** Store and manage contact information like name, phone number, and email.	02

References:

Text Books:

1. Elisabeth Robson and Eric Freeman, Head First HTML and CSS, O’Reilly, Second Edition.
2. Alex Banks and Eve Porcello, Learning React: Modern Patterns for Developing React Apps, O’Reilly, Second Edition

Reference Books:

3. Achyut Godbole and Atul Kahate, Web Technologies, Tata McGraw Hill, Third Edition (or latest).
4. Anthony Accomazzo, Ari Lerner, and Murray Nathaniel, Fullstack React: The Complete Guide to ReactJS and Friends, Fullstack.io, Latest Edition.

Data Structures Laboratory

AM0308	PCC	0L - 0T - 2P	1 Credits

Teaching Scheme	Examination Scheme
Practical: 2hrs/week	CA: 50 Marks, MSE:, ESE: 50 Marks

Pre requisites: Basic knowledge of C programming, Knowledge of basic mathematical concepts

Couse Description:

This course introduces the fundamental concepts of data structures and algorithms for efficient data organization and processing. It covers stacks, queues, linked lists, searching, sorting, hashing, trees, and graphs. Students will learn various data representation techniques and operations to solve computational problems effectively.

Course Objectives:

1	To understand the fundamental concepts of data structures, Abstract Data Types (ADTs), and algorithms.
2	To learn the implementation and applications of stack and queue data structures using different techniques.
3	To understand linked list data structures and perform various operations on singly, doubly, and circular linked lists.
4	To study searching, sorting, and hashing techniques for efficient data organization and retrieval.
5	To learn tree data structures, binary tree traversal methods, and Binary Search Tree operations.
6	To understand graph representations and graph traversal algorithms such as BFS and DFS.

Course Outcomes: On completion of the course, students will be able to,

CO1	**Apply** basic data structure concepts, ADTs, algorithms, and operations.
CO2	**Apply** stack and queue ADTs using static, dynamic, circular, and priority implementations.
CO3	**Apply** singly, doubly, and circular linked lists to perform basic operations.
CO4	**Analyze** various searching and sorting algorithms and hashing techniques, including hash functions and collision resolution methods.
CO5	**Implement** tree data structures including binary trees, their representations, traversal techniques, and operations in Binary Search Trees such as insertion and deletion.
CO6	**Analyze** graph representations and traversal techniques such as adjacency matrix, adjacency list, BFS, and DFS.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	1	0	1	0	0	0	0	0	1	1	0	0
CO2	3	2	2	1	2	0	0	0	0	0	1	2	1	0
CO3	3	2	2	2	2	0	0	0	0	0	1	2	1	0
CO4	3	3	2	2	2	0	0	0	0	0	1	2	2	1
CO5	3	2	3	2	3	0	0	0	0	0	1	3	2	2
CO6	3	3	2	2	2	0	0	0	0	0	2	3	2	1

Practical No.	**List of Practical's**	**Hrs.**
1	Implement a student record management system using arrays, functions, pointers, and structures for storing and displaying student details.	**02**
2	Implement a singly linked list to manage employee records with operations such as insertion, deletion, and searching.	**02**
3	Implement a doubly linked list for managing a music playlist with forward and backward navigation.	**02**
4	Implement a circular linked list to simulate round robin scheduling for tournament teams.	**02**
5	Implement stack operations using arrays and linked lists to simulate browser backtracking functionality.	**02**
6	Implement queue operations using arrays and linked lists for managing ticket reservation requests.	**02**
7	Implement circular queue and dequeue to simulate call center request handling and priority customer support.	**02**
8	Implement searching algorithms to search books efficiently in a library management system.	**02**
9	Implement basic sorting algorithms to arrange student marks in ascending and descending order.	**02**
10	Implement advanced sorting algorithms for efficient processing of employee salary records.	**02**
11	Implement Binary Search Tree operations such as insertion, deletion, and traversal for patient records.	**02**
12	Implement graph representation using adjacency list and perform BFS and DFS traversals for social network connections.	**02**

Python Programming Laboratory

AM0408	PCC	1L - 0T - 4P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 1hrs/week	CA: 50 Marks, MSE: - , ESE: 50 Marks

Pre requisites:
Familiarity with fundamental programming concepts such as variables, data types, operators, and control

Course Description:
This course provides practical exposure to Python programming through problem-solving and application-based laboratory exercises. It covers Python fundamentals, control structures, strings, data structures, functions, modules, file handling, and exception handling. The course emphasizes hands-on learning, logical thinking, debugging skills, and the development of simple real-world applications using Python. It serves as a foundation for advanced computing, AI, and Machine Learning courses.

Course Objectives:	
1	To introduce students to the fundamentals of Python programming including syntax, variables, data types, operators, and basic input/output operations.
2	To develop problem-solving skills by applying control statements, strings, and basic programming constructs in Python.
3	To familiarize students with Python data structures such as lists, tuples, and dictionaries for efficient data organization and manipulation.
4	To enable students to design modular programs using functions and modules for structured and reusable code development.
5	Equip students with essential programming skills using modules, file handling, and exception handling to write simple, modular, and error-resistant programs.

Course Outcomes: After completion of this course, students should be able to,	
CO1	**Remember** fundamental concepts of Python programming including syntax, variables, identifiers, data types, operators, and basic input/output statements.
CO2	**Explain** the working of control statements, string operations, and Python data structures such as lists, tuples, and dictionaries.
CO3	**Apply** Python programming constructs, data structures, functions, and file handling techniques to develop simple programs.
CO4	**Analyze** the program logic to identify errors, apply exception handling, and organize programs using modules for improved readability and reliability.

CO - PO Mapping														
Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	1	0	0	2	0	0	0	0	0	1	1	0	0
CO2	3	2	1	0	2	0	0	1	0	0	1	2	1	0
CO3	3	2	2	1	3	0	0	1	1	0	2	3	2	1
CO4	2	3	2	2	2	0	0	1	1	0	2	2	2	1

Unit No.	**Course Contents**	**Hrs.**
1	**Python Fundamentals:** Introduction to Python and applications, Python syntax and indentation, Variables, identifiers, and keywords, Basic data types: int, float, Boolean, string, Input and Output using print() and input, Basic operators: arithmetic, relational, logical.	03
2	**Control Statements and Strings:** Conditional statements: if, if-else, nested if, Looping statements: for, while, Loop control statements: break, continue, Strings: Creating strings, Indexing and slicing, Basic string operations.	03
3	**Lists, Tuples, Dictionaries:** Lists: list operations, list slices, list methods, list loop, mutability, aliasing, cloning lists, list parameters, list comprehension; Tuples: creation and indexing, tuple comprehension; Dictionaries: storing and accessing data	03
4	**Functions, Files and Exception Handling:** Functions: Function definition, Function calls, Parameters and arguments, Return values. File handling: read and write files, Introduction to modules (math, random), Creating user-defined modules. Exception handling using try–except	03

References:

Text Book:

1. Mark Lutz, Programming Python, O’Reilly Media, First Edition, ISBN No. 978-0596000851.
2. Allen B. Downey, Think Python, O’Reilly Media, First Edition, ISBN No. 978-1491939369..
3. Ch. Satyanarayana, M. Radhika Mani, B. N. Jagadesh, Python Programming, IBP Books, First Edition, ISBN No. 9789386235633.

Reference Book:

1. Wesley J. Chun, Core Python Programming, Prentice Hall, Second Edition, ISBN No. 0-13-226993-7.
2. David M. Beazley, Python Essential Reference, New Riders Publishing, First Edition, ISBN No. 978-0735709010.
3. Martin C. Brown, Python: The Complete Reference, McGraw Hill Education, First Edition, ISBN-10 9789387572942.

Practical No.	List Of Practical's	Hrs.
1	Set up Python environment and create a "Student Introduction System" that accepts student details and displays formatted information. ● Python installation, print(), input(), variables, syntax	02
2	Develop a "Simple Billing Calculator" for a grocery shop to calculate total bill, discount, and final payable amount. ● Arithmetic operators, relational operators, logical operators, input/output	02
3	Develop an "Eligibility Checker System" to determine admission eligibility based on marks and age criteria. ● if, if–else, nested if, decision-making	02
4	Create a **Traffic Signal Simulation System** using conditional logic to display actions based on signal color. ● Conditional statements, logical operations	02
5	Develop a **Monthly Expense Tracker** that continuously records and calculates expenses entered by users. ● for loop, while loop, accumulation logic	02
6	Generate **invoice and receipt patterns** using star and number pattern programs. ● Nested loops, pattern generation	02
7	Develop a **Text Analyzer Tool** to count words, vowels, spaces, and characters in a paragraph. ● Strings, indexing, slicing, string traversal	02
8	Create a **Customer Feedback Formatter** that formats feedback messages professionally. ● String formatting, built-in string methods	02
9	Develop a **Student Marks Management System** using lists for storing and updating marks. ● Lists, insertion, deletion, traversal	02
10	Create a **Product Inventory System** to search, sort, and manage product prices. ● List methods, sorting, searching	02
11	Develop a **Bus Seat Reservation Layout System** using nested lists. ● Nested lists, traversal, indexing	02
12	Develop a **Location Coordinate Processing System** for storing and manipulating fixed GPS coordinates. ● Tuples, indexing, tuple operations	02
13	Create a **Library Book Record System** to store and update book details. ● Dictionaries, key-value pairs, dictionary operations	02
14	Develop a **Phonebook / Word Frequency Counter Application** for storing contacts or analyzing text frequency. ● Dictionaries, string processing	02
15	Design a **Scientific Calculator Utility** using user-defined functions. ● Functions, parameter passing, return values	02
16	Create a **Banking Menu System** using modular functions for deposit, withdrawal, and balance inquiry. ● Functions, modular programming	02
17	Develop a **Random Password Generator / Dice Simulator** using built-in Python modules. ● Modules, math module, random module	02
18	Create a **Student Record File System** to store and retrieve student details from files. ● File handling, file modes, read/write operations	02

19	Develop a **Secure Login System** with proper error handling and invalid input detection. ● Exception handling, try-except, validation logic	02
20	Develop an **Online Quiz System** that asks multiple-choice questions, checks answers, and displays the final score. ● Conditional statements, loops, lists, functions	02
21	Create a **Simple Attendance Management System** to record and display student attendance details. • Lists, dictionaries, file handling	02
22	**Mini Project:** Develop a real-world application such as **Student Management System / Expense Tracker / Library Management System**. • Integrating functions, data structures, lists, dictionaries, file handling, exception handling, modules	06

Fundamental of AI Laboratory

AM0310	PCC	0L - 0T - 2P	1 Credits

Teaching Scheme	Examination Scheme
Practicals: 2hrs/week	CA: 50 Marks, MSE: 00 Marks, ESE: 50 Marks

Pre requisites: Basic of computing and fundamental knowledge of problem solving techniques.

Course Description: This laboratory course provides hands-on experience in implementing fundamental Artificial Intelligence concepts using programming tools and frameworks. Students will develop and test AI-based solutions involving search techniques, knowledge representation, machine learning, and intelligent agents. The course emphasizes practical problem-solving through experimentation with real-world datasets and AI algorithms.

Course Objectives:	
1	To Study of Lisp/ PROLOG.
2	**To Understand** the fundamental concepts of Artificial Intelligence, AI languages (Lisp/Prolog), and real-world AI applications.
3	**To Apply** various search algorithms such as DFS, BFS, Best First Search, and Min-Max for solving computational problems.
4	**To Design** intelligent systems using advanced AI techniques like CSP, Alpha-Beta pruning, and optimization problems.

Course Outcomes: On completion of the course, students should be able to,	
CO1	Apply AI concepts, programming techniques using Lisp/Prolog, and analyze existing AI applications.
CO2	Apply various search algorithms such as DFS, BFS, Best First Search, Min-Max, and Alpha-Beta pruning to solve AI problems.
CO3	Apply AI techniques to solve problems like 8-puzzle, 8-queens, Traveling Salesman, and map coloring using CSP.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	1	1	2	0	0	0	1	0	1	3	2	1	2
CO2	3	2	2	3	0	0	1	1	0	1	3	3	2	3
CO3	3	3	2	3	1	0	1	1	0	1	3	3	3	3
CO4	2	1	1	2	0	0	0	1	0	1	3	2	1	2
CO5	3	2	2	3	0	0	1	1	0	1	3	3	2	3
CO6	3	3	2	3	1	0	1	1	0	1	3	3	3	3

Practical No.	Practical List	Hrs.
1	Study of Lisp/ PROLOG.	2
2	Existing AI Application (e.g. Recommendation system, Carpooling, OTT channels etc.)	2
3	Solve any problem using depth first search	4
4	Solve any problem using breadth first search.	4
5	Solve an 8-puzzle problem using the best first search.	2
6	Write a program to solve Tic-Tac-Toe using Min-Max search.	2
7	Solve traveling salesman problems.	2
8	Write a program for Alpha–Beta Pruning.	2
9	Write a program to solve 8 queens problems	2
10	Write a program to solve map coloring problems using CSP.	2

Probability and Statistics

AM0401	BSC	3L -0T - 0P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Prerequisites:
Probability techniques and Basic knowledge of data handling

Course Description: This course is a foundational course designed to bridge mathematical theory with computing applications. It equips students with the tools to model uncertainty, analyze complex data, and evaluate algorithm performance. This course provides the exact mathematical backbone required for modern domains like Artificial Intelligence (AI), Machine Learning (ML), and Data Science**.**

Course Objectives:	
1	To develop problem-solving skills in calculating probabilities of events.
2	To understand random variables and their classification.
3	To introduce common probability distributions and their properties
4	To understand types and significance of correlation.
5	To teach application and interpretation of regression analysis.
6	To develop skills in using least square principle for curve fitting.

Course Outcomes: On completion of the course, students will be able to,	
CO1	Solve problems using probability axioms, rules, and theorems
CO2	Define random variables and compute their expectations, variance, and moments.
CO3	Analyze data using standard distributions such as Binomial, Poisson, and Normal.
CO4	Interpret correlation coefficients for data analysis.
C05	Evaluate and interpret regression coefficients and model fit.
C06	Apply least square method to fit curves to data points.

CO - PO Mapping														
Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	–	1	–	–	–	–	–	–	2	2	1	-
CO2	3	2	–	1	–	–	–	–	–	–	2	2	1	-
CO3	3	3	1	2	1	–	–	–	–	–	2	3	2	1
CO4	2	3	1	2	1	–	–	–	–	–	2	2	2	1
CO5	2	3	1	3	1	–	–	–	–	–	2	2	1	-
CO6	2	3	1	3	1	–	–	–	–	–	2	2	1	-

Unit No.	Course Contents	Hrs
1	**Probability Theory** Definition of probability: classical, empirical, and axiomatic approach of probability, Addition theorem of probability, Multiplication theorem of probability.	6
2	**Random Variable and Mathematical Expectation:** Random variables, Probability distributions, Probability mass function, Probability density function, Mathematical expectation, Join and marginal probability distributions.	6
3	**Theoretical Probability Distributions:** Binomial distribution, Poisson distribution, Normal distribution, Fitting of binomial distributions, Properties of binomial, Poisson, and normal distributions, Relation between binomial and normal distributions.	6
4	**Correlation:** Introduction, Types of correlation, Correlation and causation, Methods of studying correlation, Karl Pearson's correlation coefficient, Spearman's rank correlation, Coefficient, Properties of Karl Pearson's correlation coefficient and Spearman's rank correlation coefficient, Probable errors.	6
5	**Linear Regression Analysis:** Introduction, Linear and non-linear regression, Lines of regression, Derivation of regression lines of y on x and x on y, Angle between the regression lines, Coefficients of regression, Theorems on regression coefficient, Properties of regression coefficient.	6
6	**Curve Fitting and Principle of Least Square :** Fitting of the curve of the type $y=ab^x$or $y=ax^b$, fitting of straight line, fitting of second degree parabolic curve.	6

References-

Text Books

1. Fundamentals of Statistics,S. C. GuptAHimalaya Publishing House, 7th Revised and Enlarged Edition,2016.

2 Probability and Random Processes,G. V. Kumbhojkar,C. Jamnadas and Co., 14th Edition,2010.

3 Advanced Engineering Mathematics,Erwin Kreyszigth Edition, John Wiley & Sons,2006.

Reference Books

1. Engineering Mathematics,Veerarajan T,Tata McGraw-Hill, New Delhi,2010.

Database Management System

AM0402	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	**Examination Scheme**
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic programming, data structures, and fundamental concepts of logic, sets, and computer

Couse Description: This course provides a comprehensive understanding of database concepts, data modeling, relational databases, SQL, normalization, and indexing techniques. It enables students to design, secure, and manage efficient database systems with reliable storage and recovery mechanisms.

Course Objectives:

1	To Apply database system concepts to design data models using E-R diagrams and extended features
2	To Apply relational model concepts and SQL for database creation, queries, and management.
3	To Analyze database designs using functional dependencies and normalization up to BCNF.
4	To Apply storage and file organization techniques and analyze indexing and hashing methods for efficient data access.
5	To Evaluate database security issues and recommend appropriate access control mechanisms.
6	To Analyze and compare recovery methods such as shadow paging and crash recovery.

Course Outcomes: On completion of the course, students will be able to,

CO1	**Explain** database concepts, architecture, users, and E-R modeling including keys, constraints, and extended features.
CO2	**Apply** relational model concepts and SQL commands, including queries, joins, views, and data manipulation.
CO3	**Analyze** relational database designs using functional dependencies, referential integrity, and normalization (1NF to BCNF).
CO4	**Apply** data storage, file organization, indexing, and hashing techniques, including B+ trees and SQL-based indexes.
CO5	**Evaluate** database security issues and implement access control mechanisms like discretionary, mandatory, and role-based security.
CO6	Compare and implement recovery techniques such as shadow paging and crash recovery.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	-	-	-	-	2	-	1	1	-	-
CO2	3	3	2	2	3	-	-	-	2	1	2	2	2	1
CO3	3	3	3	2	-	-	-	-	-	-	2	2	1	1
CO4	3	2	2	2	3	-	-	-	-	-	2	2	2	2

CO5	2	2	2	-	2	3	3	-	-	1	2	1	1	1
CO6	3	3	2	2	2	-	-	-	-	-	2	2	2	1

Unit No.	Course Contents	Hrs.
1	**Unit 1: Introduction to databases and E-R model** : Purpose of Database Systems, View of data, Database architecture, Database users and administrator, E-R model: Entity sets, Relationship sets, Mapping Constraints, Keys, E-R Diagram, Reducing E-R Diagrams to relational schemas, Extended E-R features: Specialization, Generalization, and Aggregation	**06**
2	**Unit 2: Relational Model and SQL Relational Model:** Structure of Relational Database, Reduction of ER model into Relational schemas, Schema-instance distinction, Referential integrity and foreign keys, pure languages, Relational algebr4 Example queries SQL: Introduction to SQL, Data definition statements with constraints, Insert, Update and Delete, Set operations, Aggregate functions group by and having clauses, Nested Queries, Views, Complex Queries, Joins.	**07**
3	**Unit 3: Relational Database**: Design Referential Integrity, features of good relational designs, functional dependency, closure of a set of functional dependencies and Canonical cover. **Normalization**: Purpose of normalization, First Normal Form (1NF), Second Normal Form (2NF), Third Normal Form (3NF), Boyce-Codd Normal Form (BCNF).	**06**
4	**Unit 4: Data Storage and Indexing Storage and File structure:** Overview of physical storage media, RAID, File Organization, Organization of Records in Files, Data Dictionary Storage, Database Buffer. **Indexing and Hashing**: Basic Concepts, Ordered Indices, B+ Tree Index Files, Multiple Key Access. Static Hashing, Dynamic Hashing, Index definition in SQL	**07**
5	**Unit 5: Database Security and Authorization:** Introduction to Database Security Issues, Discretionary Access Control Based on Granting and Revoking Privileges, Mandatory Access Control and Role-Based Access Control for Multilevel Security, Introduction to Statistical Database Security	**05**
6	**Unit 6: Recovery System** Failure classification, Storage structure, Implementation of stable and Atomicity, Log based recovery, Checkpoints, Shadow paging, crash recovery**.**	**05**

Text Book:

1. Database system concepts, A. Silberschatz, H.F. Korth, S. Sudarshan, McGraw Hill Education, 2011
2. Database Systems- A practical approach to Design, Implementation, Thomos Connolly, Carolyn Begg, Pearson Education, 2009
3. Database Systems - Design, Irnplententation and Managernent, Rob & Coronel, Thomson Course Technology, 2008
4. Database Management Systems, Raghu Ram Krishnan, McGraw Hill, 2002

Reference Book:

1. Fundamentals of Database Systems, RamezElmasri and ShamkantNavathe, Pearson Education, 2007
2. Database Systems: Design implementation and management, Peter Rof. Carlos Coronel, Cengage Learning, 2014.

Design Analysis and Algorithm

AM0403	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	**Examination Scheme**
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic Programming Knowledge and Data Structure

Course Description: This course focuses on the design, analysis, and optimization of algorithms for solving computational problems efficiently. Students will learn various algorithmic strategies such as divide and conquer, greedy methods, dynamic programming, backtracking, and branch and bound. The course emphasizes the analysis of time and space complexity using asymptotic notations. Learners will develop skills to compare and evaluate different algorithms for performance and scalability

Course Objectives:

1	To explain fundamental algorithm design, analysis, and performance evaluation concepts using asymptotic notations and recurrence solving techniques.
2	To implement algorithms for searching, sorting, shortest path, matrix operations, and combination optimization, and evaluate their time and space complexity
3	To solve computational Problems by applying algorithmic paradigms greedy.
4	To compare algorithmic strategies for problem-solving and justify the choice of technique based on efficiency and problem constraints.
5	To develop the ability to design efficient algorithmic solutions for optimization problems
6	To classify problems into complexity classes P, NP, and NP-complete, and apply polynomial-time reductions to analyse computational hardness.

Course Outcomes: On completion of the course, students Should be able to,

CO1	Analyze time and space complexities of algorithms.
CO2	Identify algorithm design methodology to solve problems.
CO3	Apply greedy strategy to solve optimization problems
CO4	Develop solutions using backtracking and branch-and-bound techniques
CO5	Apply Dynamic Programming techniques to formulate and solve optimization problems
CO6	Distinguish between P and NP classes of problems

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	3		2								2	1	0
CO2	3	3	2									2	1	0
CO3	2	3	3									3	2	1

CO4	2	3	3	2								3	2	2
CO5	3	3	3	2								3	3	2
CO6	3	3		2								2	1	1

Unit No.	Course Contents	Hrs.
1	**Introduction to Algorithms:** Definition, Properties of Algorithms, Expressing Algorithm, Algorithm Design Techniques, Performance Analysis of Algorithms, Types of Algorithms Analysis, Order of Growth, Asymptotic Notations, Recursion, Recurrences Relation, Substitution Method, Iterative Method, Recursion Tree, Master Theorem.	6
2	**Divide and Conquer:** Introduction, Binary Search, Merge Sort, Quick Sort, Strassen‘s Matrix Multiplication.	6
3	**Greedy Algorithms:** General Method, Optimal Merge Patterns, Huffman Coding, Knapsack Problem, Activity Selection Problem, Job Sequencing with Deadline, Minimum Spanning Tree, Single-Source Shortest Path Algorithm.	6
4	**Backtracking:** Backtracking Concept, N–Queens Problem, Four–Queens Problem, Eight–Queen Problem, Hamiltonian Cycle, Sum of Subsets Problem, Graph Coloring Problem, Branch and Bound: Introduction, 15-Puzzle Problem, Comparisons between Backtracking and Branch and Bound.	6
5	**Dynamic Programming**: Introduction, Characteristics of Dynamic Programming, Component of Dynamic Programming, Comparison of Divide-and-Conquer and Dynamic Programming Techniques, Longest Common Subsequence, matrix multiplication, shortest paths: Bellman-Ford, Floyd Warshall, Application of Dynamic Programming.	7
6	**NP Completeness**: Introduction, the Complexity Class P, the Complexity Class NP, Polynomial-Time Reduction, the Complexity Class NP-Complete, Traveling Salesperson Problem, Cook’s Theorem	5

References:

Text Book:

1. Fundamentals of Computer Algorithms, Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, Universities Press, 2nd 2011.
2. Introduction to Algorithms, T. Cormen, PHI Publication, 2nd 2022.
3. Data Structure and Algorithms, Aho, Ullman, Addison-Wesley Publication, 1st Edition, 1983.

Reference Book:

1. Foundation of Algorithm, Richard E. Neapolitan, Jones & Bartlett Learning Learning, 5th 2011.
2. Design and analysis of algorithms, Mohan, I. Chandra, PHI Publication, 2nd 2010.

Cloud computing

AM0404	PCC	3L - 1T - 0P	3 Credits

Teaching Scheme	**Examination Scheme**
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites -1.Knowledge of operating systems and computer networks
2.Basic architecture of client-server model and distributed computing

Course Discription:_ This course provides a comprehensive introduction to the concepts, architectures, technologies, and applications of cloud computing. It covers the fundamental principles of cloud service models (IaaS, PaaS, and SaaS), deployment models (Public, Private, Hybrid, and Community Clouds), and virtualization technologies that form the foundation of cloud environments. Students will explore cloud infrastructure, resource

Course Objectives:	
1	To understand the fundamental concepts, characteristics, and reference models of cloud computing along with its advantages, limitations, and emerging trends.
2	To study cloud service models, deployment architectures, cloud storage mechanisms, and AI-based cloud frameworks used in modern computing environments.
3	To explore virtualization technologies, their classifications, implementation techniques, and role in cloud computing infrastructure.
4	To examine major cloud platforms, server less computing, edge AI technologies, and their applications in real-world cloud ecosystems.
5	To develop the ability to evaluate and apply cloud computing solutions for scientific, business, and enterprise applications with consideration of scalability, efficiency, and security
6	To analyze cloud security architectures, advanced cloud applications, containerization, orchestration, and emerging topics in cloud computing.

Course Outcomes: On completion of the course, students will be able to,	
CO1	**Illustrate** the working of cloud computing models.
CO2	**Compare** the architectures, service & deployment models of cloud computing.
CO3	**Explore** the context of virtualization technologies and its need.
CO4	**Examine** cloud computing platforms and its underlying security architecture.
CO5	**Examine** general issues in the cloud computing.
CO6	**Elaborate** advanced applications of cloud computing in different sectors.

CO - PO Mapping														
Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	-	-	2	-	-	-	1	-	2	2	1	-
CO2	3	3	2	1	2	1	-	-	1	-	2	2	1	1
CO3	2	2	2	1	3	2	-	-	-	-	2	2	2	1
CO4	3	2	2	2	3	-	1	1	1	-	2	3	2	2
CO5	2	3	1	2	2	3	2	-	1	-	2	1	1	1
CO6	2	2	3	2	3	2	1	2	2	1	3	3	3	3

Unit No.	Course Contents	Hrs.
1	**Basics of Cloud Computing** Introduction and Characteristics of Cloud Computing, Cloud computing reference model, Advantages and Disadvantages of cloud computing, Challenges ahead, Edge computing.	**06**
2	**Models, Architecture and Storage** NIST architecture, Delivery Models: IaaS, PaaS, SaaS, MLaaS Deployment models, Cloud storage, Cloud-Based AI Frameworks and Libraries, Open challenges.	**06**
3	**Virtualization** Characteristics of virtualized environments, Taxonomy of virtualization techniques, Virtualization and cloud computing, Pros and cons of virtualization, Paravirtualization and Full virtualization.	**06**
4	**Cloud Platforms** Amazon web services, Google AppEngine, Microsoft Azure, Aneka platform, Edge AI & Serverless Computing.	**06**
5	**Cloud Computing Security Architecture** Architectural Considerations, General Issues, Trusted Cloud Computing, Secure Execution environments and Communications, Identity Management and Access Control, AI driven cloud security.	**06**
6	**Cloud Applications and Advanced Topics** Scientific applications, Business and consumer applications, Energy efficiency in clouds, Serverless Computing (FaaS), Containerization and Orchestration.	**06**

References

Text Books:

- Rajkumar Buyya, Australia Christian Vecchiola, Australia S. Thamarai Selvi, “Mastering Cloud Computing Foundations and Applications Programming”, Elsevier Inc. John W.

Reference Books:

- Rajkumar Buyya, James Broberg, Andrzej Goscinski, “Cloud Computing: Principles and Paradigms”, John Wiley & Sons, Inc.
- Judith Hurwitz, R. Bloor, M. Kaufman, F. Halper, “Cloud Computing for Dummies”, Wiley India Edition.
- John W. Rittinghouse, James F. Ransome, “Cloud Computing Implementation, Management and Security”, CRC Press.

Tutorial No.	**Tutorial List**
1.	**Introduction to Cloud Computing** **Objective:** Understand the fundamentals of cloud computing. **Tasks:** 1.Define cloud computing. 2.Explain characteristics of cloud computing. 3.Compare traditional computing and cloud computing. 4.Identify real-world cloud services used daily.
2.	**Cloud Computing Reference Model** **Objective:** Study cloud architecture and reference model. **Tasks:** 1.Draw and explain the cloud computing reference model. 2.Identify cloud consumers, providers, brokers, auditors, and carriers.
3.	**Advantages, Disadvantages and Challenges of Cloud Computing** **Objective**: Analyze cloud adoption factors. **Tasks:** **1.**List advantages and disadvantages of cloud computing. 2.Discuss challenges such as security, privacy, latency, and interoperability.
4.	**Cloud Service Models (IaaS, PaaS, SaaS, MLaaS)** **Objective**: Compare cloud delivery models**.** Tasks: 1**.**Explain IaaS, PaaS, SaaS, and MLaaS. . 2.Identify providers and services for each model. . . 3.Create a comparison table
5.	**Cloud Deployment Models and Storage** **Objective**: Understand cloud deployment options. **Tasks: 1.**Compare Public, Private, Hybrid, and Community Clouds. . . 2.Study cloud storage architecture. . . . 3.Explore object, block, and file storage.
6.	**Virtualization Fundamentals** **Objective**: Learn virtualization concepts**.** **Tasks:** 1.Define virtualization. . . . 2.Explain characteristics of virtualized environments. . 3.Study hypervisors and virtual machines.

7.	**Amazon Web Services (AWS)** **Objective:** Explore AWS cloud platform. **Tasks:** 1.Create an AWS account. . 2.Study EC2, S3, Lambda, and RDS services. . 3.Launch a virtual machine on EC2.
8.	**Microsoft Azure and Google App Engine** **Objective:** Study major cloud platforms**.** **Tasks:** 1.Explore Azure services and architecture. . 2.Study Google App Engine deployment. . . 3.Compare AWS, Azure, and Google Cloud.
9.	**Cloud Security Architecture** **Objective:** Understand cloud security mechanisms**.** **Tasks**: 1**.**Study cloud security architecture. . . 2.Analyze threats and vulnerabilities. . 3. Explore trusted cloud computing concepts.
10.	**Serverless Computing,Containers and Cloud Applications** **Objective:** Explore advanced cloud technologies. **Tasks:** 1.Study Serverless Computing (FaaS). . 2.Understand Docker containerization. . 3.Explore Kubernetes orchestration.

Machine Learning

AM0405	Multi-Disciplinary Minor-02	3L - 0T - 0P	3 Credits

Teaching Scheme	**Examination Scheme**
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic python programming Language Knowledge

Course Description: This course introduces the fundamental concepts, techniques, and applications of Machine Learning for intelligent data-driven decision-making. Students will learn supervised and unsupervised learning algorithms, including regression, classification, clustering, and dimensionality reduction methods. The course covers model evaluation, feature selection, overfitting, and performance optimization techniques. Learners will gain knowledge of real-world machine learning applications in areas such as healthcare, finance, and artificial intelligence.

Course Objectives:	
1	To Fundamental concepts of Machine Learning, types of learning, data handling techniques, and real-world applications.
2	Understand and analyze various evaluation metrics used to measure the performance of classification and regression models.
3	To develop understanding of linear regression concepts and train students to build predictive models using optimization techniques like gradient descent.
4	To equip students with knowledge of classification techniques using logistic regression.
5	To provide understanding of ensemble learning and probabilistic approaches for solving classification and regression problems.
6	To familiarize students with instance-based and margin-based learning algorithms for solving complex machine learning problems.

Course Outcomes: On completion of the course, students will be able to,	
CO1	**Explain** the fundamental concepts of Machine Learning including types of learning, data representation, dataset splitting, and real-world applications.
CO2	**Analyze** the performance of machine learning models using classification and regression evaluation metrics.
CO3	**Apply** linear regression techniques and optimization methods such as gradient descent for predictive modeling.
CO4	**Apply** classification algorithms including logistic regression for solving data-driven problems.
CO5	**Evaluate** ensemble and probabilistic models such as random forest and Decision Tree for classification and regression tasks.
CO6	**Evaluate** machine learning algorithms including KNN and SVM for classification, regression, and real-world problem solving.

CO - PO Mapping														
Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	0	1	0	0	0	1	0	1	2	1	1
CO2	2	3	1	2	2	0	0	0	1	0	1	2	2	1
CO3	3	2	2	2	3	0	0	0	1	0	1	3	2	2
CO4	3	3	3	2	3	1	0	1	1	1	1	3	3	2
CO5	2	3	2	2	3	1	0	1	1	0	1	3	3	3
CO6	2	3	2	2	3	1	0	1	1	0	2	3	3	2

Unit No.	**Course Contents**	**Hrs.**
1	**Introduction to Machine Learning** Introduction to Machine Learning: Definition of Machine Learning, Definition of learning. Life Cycle of Machine learning, Classification of Machine Learning: Supervised learning, unsupervised learning, Reinforcement learning, Semi-supervised learning. Categorizing based on required Output: Classification, Regression, and Clustering. Difference in ML and Traditional Programming, Definition of Data, Information and Knowledge. Split data in Machine Learning: Training Data, Validation Data and Testing Data. Machine Learning: Applications	**7**
2	**Machine Learning - Performance Metrics** Performance Metrics for Classification Problems- Confusion Matrix, Classification Accuracy, Classification Report- Precision, Recall or Sensitivity, Support, F1 Score, AUC (Area Under ROC curve). Performance Metrics for Regression Problems- Mean Absolute Error (MAE), Mean Square Error (MSE), R Squared (R2)	**6**
3	**Linear Regression** Introduction to linear regression: Introduction to Linear Regression, Optimal Coefficients, Cost function, Coefficient of Determination, Analysis of Linear Regression using dummy Data, Linear Regression Intuition. Multivariable regression	**5**
4	**Logistic Regression** Logistic regression: Handling Classification Problems, Logistic Regression, Cost Function, Finding Optimal Values, Solving Derivatives, Multiclass Logistic Regression, Finding Complex Boundaries and Regularization, Using Logistic Regression from Sklearn.	**6**
5	**Decision Trees and Random Forests** Decision Trees, Decision Trees for Interview call, Building Decision Trees, information gain, Gain Ratio, Gini Index, Decision Trees & Overfitting, Pruning. Introduction to Random Forests, Data Bagging and Feature Selection, Extra Trees, Regression using decision Trees and Random Forest, Random Forest in Sklearn,	**6**
6	**KNN and SVM** K-Nearest Neighbours: Introduction to KNN, Feature scaling before KNN, KNN in Sklearn, Cross Validation, Finding Optimal K, Implement KNN, Support Vector Machine: Intuition behind SVM, SVM Cost Function, Decision Boundary & the C parameter, using SVM from Sklearn,	**6**

References:

Text Book:

1. Ethem Alpaydın, Introduction to Machine Learning, PHI, Third Edition, ISBN No. 978-81-203- 5078-6.
2. Tom Mitchell, Machine Learning, Mcgraw-Hill, First Edition, ISBN No. 0-07-115467-1.
3. Christopher M. Bishop, Pattern Recognition and Machine Learning, Mcgraw-Hill, ISBN No. 0- 07- 115467-1

Reference Book:

1. Shai shalev-Shwartz and Shai Ben-David, Understanding Machine Learning (From Theory to Algorithms), Cambridge University Press, First Edition, ISBN No. 978-1-107-51282-5.
2. A. Rostamizadeh, A. Talwalkar, M. Mohri, Foundations of Machine Learning, MIT Press.

Machine Learning

AM0406	PCC	3L - 0T - 0P	3 Credits

Teaching Scheme	Examination Scheme
Lectures: 3hrs/week	CA: 20 Marks, MSE: 20 Marks, ESE: 60 Marks

Pre requisites: Basic python programming Language Knowledge

Course Description: This course introduces the fundamental concepts, techniques, and applications of Machine Learning for intelligent data-driven decision-making. Students will learn supervised and unsupervised learning algorithms, including regression, classification, clustering, and dimensionality reduction methods. The course covers model evaluation, feature selection, overfitting, and performance optimization techniques. Learners will gain knowledge of real-world machine learning applications in areas such as healthcare, finance, and artificial intelligence.

Course Objectives:	
1	To fundamental concepts of Machine Learning, types of learning, data handling techniques, and real-world applications.
2	To understand and analyze various evaluation metrics used to measure the performance of classification and regression models.
3	To develop understanding of linear regression concepts and train students to build predictive models using optimization techniques like gradient descent.
4	To equip students with knowledge of classification techniques using logistic regression and decision tree algorithms.
5	To provide understanding of ensemble learning and probabilistic approaches for solving classification and regression problems.
6	To familiarize students with instance-based and margin-based learning algorithms for solving complex machine learning problems.

Course Outcomes: On completion of the course, students should be able to,	
CO1	**Explain** the fundamental concepts of Machine Learning including types of learning, data representation, dataset splitting, and real-world applications.
CO2	**Analyze** the performance of machine learning models using classification and regression evaluation metrics.
CO3	**Apply** linear regression techniques and optimization methods such as gradient descent for predictive modeling.
CO4	**Apply** classification algorithms including logistic regression and decision trees for solving data-driven problems.
CO5	**Evaluate** ensemble and probabilistic models such as random forest and Naive Bayes for classification and regression tasks.
CO6	**Evaluate** machine learning algorithms including KNN and SVM for classification, regression, and real-world problem solving.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	0	1	0	0	0	1	0	1	2	1	1
CO2	2	3	1	2	2	0	0	0	1	0	1	2	2	1
CO3	3	2	2	2	3	0	0	0	1	0	1	3	2	2
CO4	3	3	3	2	3	1	0	1	1	1	1	3	3	2
CO5	2	3	2	2	3	1	0	1	1	0	1	3	3	3
CO6	2	3	2	2	3	1	0	1	1	0	2	3	3	2

Unit No.	Course Contents	Hrs.
1	**Introduction to Machine Learning** Introduction to Machine Learning: Definition of Machine Learning, Definition of learning. Life Cycle of Machine learning, Classification of Machine Learning: Supervised learning, unsupervised learning, Reinforcement learning, Semi-supervised learning. Categorizing based on required Output: Classification, Regression, and Clustering. Difference in ML and Traditional Programming, Definition of Data, Information and Knowledge. Split data in Machine Learning: Training Data, Validation Data and Testing Data. Machine Learning: Applications	**7**
2	**Machine Learning - Performance Metrics** Performance Metrics for Classification Problems- Confusion Matrix, Classification Accuracy, Classification Report- Precision, Recall or Sensitivity, Support, F1 Score, AUC (Area Under ROC curve). Performance Metrics for Regression Problems- Mean Absolute Error (MAE), Mean Square Error (MSE), R Squared (R2)	**5**
3	**Linear Regression** Introduction to Linear Regression, Optimal Coefficients, Cost function, Coefficient of Determination, Analysis of Linear Regression using dummy Data, Linear Regression Intuition. Multivariable regression and gradient descent: Generic Gradient Descent, Learning Rate, Complexity Analysis of Normal Equation Linear Regression, How to find More Complex Boundaries, Variations of Gradient Descent.	**5**
4	**Logistic Regression and Decision Trees** Logistic regression: Handling Classification Problems, Logistic Regression, Cost Function, Finding Optimal Values, Solving Derivatives, Multiclass Logistic Regression, Finding Complex Boundaries and Regularization, Using Logistic Regression from Sklearn. Decision Trees, Decision Trees for Interview call, Building Decision Trees, Getting to Best Decision Tree, Deciding Feature to Split on, Continuous Valued Features Code using Sklearn decision tree, information gain, Gain Ratio, Gini Index, Decision Trees & Overfitting, Pruning.	**6**

5	**Random Forests and Naive Bayes** Introduction to Random Forests, Data Bagging and Feature Selection, Extra Trees, Regression using decision Trees and Random Forest, Random Forest in Sklearn, Naive Bayes: Bayes Theorem, Independence Assumption in Naive Bayes, Probability estimation for Discrete Values Features, How to handle zero probabilities, Implementation of Naive Bayes, Finding the probability for continuous valued features, Text Classification using Naive Bayes.	**6**
6	**KNN and SVM** K-Nearest Neighbours: Introduction to KNN, Feature scaling before KNN, KNN in Sklearn, Cross Validation, Finding Optimal K, Implement KNN, Curse of Dimensionality, Handling Categorical Data, Pros & Cons of KNN. Support Vector Machine: Intuition behind SVM, SVM Cost Function, Decision Boundary & the C parameter, using SVM from Sklearn, Finding Non Linear Decision Boundary, Choosing Landmark Points, Similarity Functions, How to move to new dimensions, Multi-class Classification, Using Sklearn SVM on Iris, Choosing Parameters using Grid Search, Using Support Vectors to Regression.	**7**

References:

Text Book:

1. Ethem Alpaydın, Introduction to Machine Learning, PHI, Third Edition, ISBN No. 978-81-203- 5078-6.
2. Tom Mitchell, Machine Learning, Mcgraw-Hill, First Edition, ISBN No. 0-07-115467-1.
3. Christopher M. Bishop, Pattern Recognition and Machine Learning, Mcgraw-Hill, ISBN No. 0- 07- 115467-1

Reference Book:

1. Shai shalev-Shwartz and Shai Ben-David, Understanding Machine Learning (From Theory to Algorithms), Cambridge University Press, First Edition, ISBN No. 978-1-107-51282-5.
2. A. Rostamizadeh, A. Talwalkar, M. Mohri, Foundations of Machine Learning, MIT Press.

Machine Learning Laboratory

AM0407	PCC	0L - 0T - 2P	1 Credits

Teaching Scheme	Examination Scheme
Practical: 2hrs/week	CA: 50 Marks ESE:50 Marks

Pre requisites: Python Programming

Course Description: This laboratory course provides practical exposure to the implementation of machine learning algorithms using Python and popular ML libraries. Students will perform data preprocessing, feature engineering, model training, and evaluation on real-world datasets. The course covers supervised and unsupervised learning techniques, including classification, regression, clustering, and model validation. Through hands-on experiments, learners will gain experience in developing and optimizing machine learning models.

Course Objectives:	
1	To understand and implement various Machine Learning algorithms for regression and classification problems using Python and Scikit-learn
2	To analyze and evaluate the performance of Machine Learning models using suitable datasets and performance metrics.
3	To develop practical skills in data preprocessing, model training, prediction, and result interpretation for real-world applications.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	2	2	1	1	3	0	0	0	1	0	1	2	2	1
CO2	3	2	2	1	3	0	0	0	1	0	1	3	2	1
CO3	3	3	2	2	3	1	0	1	1	0	1	3	3	2
CO4	2	3	1	3	2	0	0	0	1	0	1	2	3	2
CO5	3	3	3	2	3	1	1	1	2	1	2	3	3	3
CO6	2	2	1	1	3	0	0	0	1	0	1	2	2	1

Course Outcomes: On completion of the course, students should be able to,	
CO1	Apply data preprocessing techniques for Machine Learning datasets.
CO2	Implement regression algorithms for prediction problems.
CO3	Develop classification models using Machine Learning algorithms.
CO4	Analyze the performance of Machine Learning models using evaluation metrics.
CO5	Design Machine Learning solutions for real-world applications and mini projects.

Practical's List

Practical. No.	Practical's List	Hrs
1.	Develop a Machine Learning model to predict the salary of an employee based on years of experience using Linear Regression.	02
2.	Develop a Multiple Linear Regression model to predict house prices based on multiple features such as area, number of bedrooms, age of the house, and location score.	02
3.	Build a Logistic Regression model to predict whether a student will pass or fail an examination based on study hours and attendance percentage.	02
4.	Design a Decision Tree classifier to determine whether a customer is eligible for a bank loan based on income, credit score, age, and employment status.	02
5.	Develop a Random Forest classification model to detect fraudulent online transactions using transaction amount, payment mode, transaction location, and customer history.	02
6.	Implement an SVM classifier to classify emails as spam or non-spam based on text features extracted from email content.	02
7.	Build a Naive Bayes model for sentiment analysis to classify customer reviews as positive or negative.	02
8.	Develop a KNN-based recommendation system to classify iris flowers into different species based on petal and sepal measurements.	02
9.	Mini Project-I	04
10.	Mini Project-II	04

Java Programming Laboratory

AM0309	**PCC**	**2L - 0T - 2P**	**3 Credits**

Teaching Scheme	**Examination Scheme**		
Lectures: 2hrs/week	CA: 50 Marks,	MSE: - ,	ESE: 50 Marks

Pre requisites:
Knowledge of C Programming, Understanding of Object-Oriented Concepts using C++.

Course Description:
This course enables students to acquire practical skills in Java programming and object-oriented software development. It focuses on the implementation of programming constructs, modular application design, exception handling, concurrent programming, collections, and file management. The laboratory activities are designed to enhance analytical thinking, debugging skills, and the ability to develop real-world software solutions using industry-relevant programming practices.

Course Objectives:	
1	To introduce object-oriented programming concepts using Java.
2	To develop the ability to write structured Java programs using control statements and arrays.
3	To understand OOP features such as classes, inheritance, and interfaces.
4	To learn exception handling and multithreading mechanisms in Java.
5	To understand file handling and graphical programming concepts.

Course Outcomes: After completion of this course, students should be able to,	
CO1	Explain Java programming fundamentals, syntax, operators, and control structures.
CO2	Apply object-oriented concepts such as classes, objects, constructors, arrays, and Inheritance.
CO3	Implement modular Java programs using interfaces, packages, and exception handling Mechanisms.
CO4	Develop Java applications using multithreading and file handling concepts.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	1	0	0	1	0	0	0	0	0	1	1	0	0
CO2	3	2	2	0	2	0	0	0	0	0	1	2	1	0
CO3	3	2	2	0	2	0	0	0	1	0	1	2	2	1
CO4	3	3	2	1	3	0	0	1	1	0	2	3	2	2

Unit No.	Course Contents	Hrs.
1	**Introduction to Java Programming and Basic Programming Constructs:** Introduction to Java, Features of Java, Java vs C++, Java Virtual Machine (JVM), JRE, JDK, Java program structure, Compilation and execution of Java program, Data types, variables, constants, Type casting, Operators and expressions, Input/output using Scanner and System.out, Control statements: (if, if-else, switch), Looping (for, while, do-while).	06
2	**Object-Oriented Programming in Java:** Concepts of OOP: Classes and Objects, Data Abstraction, Encapsulation, Inheritance, Polymorphism, Method Overloading, Method Overriding, Constructors, Constructor Overloading, Static Members and Methods, Arrays and Arrays of objects,	06
3	**Interfaces, Exception Handling and Multithreading:** Interfaces: Defining Interfaces, Implementing Interfaces, Multiple inheritance using interfaces, interface vs abstract class, Exception Handling: Types of Exceptions, try, catch, finally, throw and throws, User-defined exceptions, Multithreading: thread life cycle, creating threads using Thread class and Runnable interface, thread synchronization.	07
4	**Collections and File Handling:** Java Collections: List, Set, Map, ArrayList, HashMap, File Handling: File class, Byte streams, Character streams, Reading and writing files	05

References:

Text Book:

1. Herbert Schildt, Java: The Complete Reference, McGraw-Hill, Eighth Edition, ISBN No. 978-1259002465.
2. Bruce Eckel, Thinking in Java, Prentice Hall, Fourth Edition, ISBN No. 978-0131872486.
3. E. Balagurusamy, Programming with Java: A Primer, Tata McGraw-Hill, Fourth Edition, ISBN No. 978-0070141698.

Reference Books:

4. Herbert Schildt, Java: The Complete Reference, McGraw-Hill, Twelfth Edition, ISBN No. 978-1260463415.
5. K. Somasundaram, Programming in JAVA, LAP Lambert Academic Publishing, First Edition, ISBN No. 978-3330081260.

Practical No.	List Of Practical's	Hrs.
1	Develop a **Simple Billing System** for a grocery shop that accepts product details and calculates the total bill using arithmetic and logical operations. ● Java syntax, Data types, Variables, Operators, Output statements	02
2	Develop a **Student Grade Evaluation System** that assigns grades based on marks using conditional statements. ● if-else, nested if, switch statements	02
3	Develop an **ATM Transaction Simulation** that repeatedly displays banking operations until the user exits the system. ● for loop, while loop, do-while loop	02
4	Develop a **Student Marks Analysis System** to store marks of students in arrays and calculate highest, lowest, and average marks. ● Arrays, Array operations, Looping	02
5	Develop an **Employee Information Management System** using classes and objects to store and display employee records. ● Classes, Objects, Methods	02
6	Develop a **Bank Account Management System** demonstrating different ways of initializing account details using constructor overloading. ● Constructors, Constructor Overloading	02
7	Develop a **Library Management System** where different types of users (Student, Faculty) inherit common properties from a base class and override methods. ● Inheritance, Method Overriding, super keyword	02
8	Develop a **Smart Device Control System** where multiple devices implement common functionalities using interfaces. ● Interfaces, Multiple Inheritance, Exception Handling	02
9	Develop a **Railway Reservation Simulation** using multithreading and collections. ● Multithreading, ArrayList / HashMap	02
10	**Mini Project** Student Record Management System . Develop a **menu-driven Java application** to manage student records with the following functionalities: ● Add student record ● Search student record ● Update student details ● Delete student record ● Display all records ● Store and retrieve records using files ● Handle invalid inputs using exception handling	06

Database Management System Laboratory

AM0409	**PCC**	**0L - 0T - 2P**	**1 Credits**

Teaching Scheme	**Examination Scheme**
Practical: 2hrs/week	CA: 50 Marks, ESE: 50 Marks

Pre requisites: Basic programming, data structures, and fundamental concepts of logic, sets, and computer organization.

Couse Description: This course provides practical knowledge of database design, SQL, PL/SQL, database connectivity, and hashing techniques for efficient data management and retrieval. It enables students to design

Course Objectives:	
1	To Apply database system concepts to design data models using E-R diagrams and extended features.
2	To Apply relational model concepts and SQL for database creation, queries, and management.
3	To Analyze database designs using functional dependencies and normalization up to BCNF.
4	To Apply storage and file organization techniques and implement indexing and hashing methods.
5	To Evaluate database security issues and suggest access control mechanisms.

Course Outcomes: On completion of the course, students will be able to,	
CO1	**Sketch** E-R diagram for given Case Study/ Problem Statement.
CO2	**Design** relational database using Normalization and Functional Dependency.
CO3	**Implement** SQL query for various operations like retrieval, insertion and manipulation of data etc.
CO4	**Implement** PL/SQL cursor, procedure, function and trigger
CO5	**Apply** hashing mechanism to build hash index file on given records.
CO6	**Develop** a program to connect database to an application program

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	1	-	-	-	1	-	-	2	1	1
CO2	3	3	3	2	2	-	-	-	1	-	-	2	2	1
CO3	3	2	3	2	3	-	-	-	1	-	1	2	3	2
CO4	3	2	3	2	3	-	-	-	1	-	1	2	3	2
CO5	3	3	2	2	3	-	-	-	-	-	1	3	2	2
CO6	3	2	3	2	3	1	-	1	2	2	2	3	3	3

Course Contents:		
Sr.No	**Title Of Experiments:**	**Hours**
1	ER Diagram of an Organization - Draw an E-R Diagram for any organization like Insurance Company, Library systems, College Management systems, Hospital Management systems etc. Use data modeling tools like Oracle SQL Developer to draw an ER diagram.	02
2	Conversion of ER Diagram to Tables- Convert the above-mentioned E-R Diagram in Relational Tables	02
3	DDL Statements – Execute DDL commands to create, alter, rename, truncate and drop tables in SQL. Apply all types of constraints, such as primary key, foreign key, not null, unique, and check.	02
4	DML Statements – Use DML Queries to insert, delete, update & display records of the tables	02
5	SQL character functions, String functions – Display the results using String operations.	02
6	Aggregate functions – Display the records using Aggregate functions and Group by, having, between, Order by clauses.	02
7	Join operations and set operations – Display the results of union, intersection, set difference, Cartesian product and Join operations of two different tables.	02
8	Views, Sub queries –Create Views for the table. Solve sub queries for given questions.	02
9	Demonstrate PLSQL Functions and Procedures.	02
10	Demonstrate Cursors, and triggers using PL/SQL.	02
11	Database Connectivity – Write a program of Database connectivity with any object oriented language.	02
12	Write a program to implement dynamic hashing and demonstrate how records are inserted, deleted, and searched efficiently.	02

References:-

Text Book:

1. Database System Concepts by A. Silberschatz, H.F. Korth, S. Sudarshan, McGraw Hill Education, 2011.
2. Database Systems – A Practical Approach to Design, Implementation by Thomas Connolly, Carolyn Begg, Pearson Education, 2009.
3. Database Systems – Design, Implementation and Management by Rob & Coronel, Thomson Course Technology, 2008.

Reference Book:

1. Fundamentals of Database Systems by Ramez Elmasri and Shamkant Navathe, Pearson Education, 2007.
2. Database Systems: Design Implementation and Management by Peter Rob, Carlos Coronel, Cengage Learning, 2014.

Mobile Application Development Laboratory

AM0410	VSEC	0L - 0T - 2P	2 Credits

Teaching Scheme	Examination Scheme
Practical: 2hrs/week	CA: 50 Marks, MSE: - ESE: - Total: 50 Marks

Pre requisites:
Basic HTML, CSS, JavaScript programming concepts.

Couse Description:
This course introduces modern mobile and web application development using Flutter, Flask API, Django, and MongoDB. It focuses on building responsive full-stack applications with authentication, API integration, and deployment using modern technologies.

Course Objectives:

1	To Introduce students to modern mobile and web application development tools and environments.
2	To Develop responsive mobile application interfaces using Flutter framework.
3	To Build RESTful backend APIs using Flask API and Django frameworks.
4	To Integrate MongoDB database with mobile and web applications.
5	To Implement authentication, API communication, and cloud-based services in applications.
6	To Deploy full-stack real-time applications using modern development technologies.

Course Outcomes: After Completion of the course students should able to,

CO1	**Configure** modern mobile and web application development environments and tools.
CO2	**Design** responsive user interfaces for mobile applications using Flutter widgets and navigation components.
CO3	**Develop** RESTful APIs and backend services using Flask API and Django frameworks.
CO4	**Integrate** MongoDB databases with applications and perform CRUD operations efficiently.
CO5	**Implement** API integration, authentication mechanisms, and cloud-based services in applications.
CO6	**Deploy** full-stack real-time applications using Flutter, Flask API/Django, and MongoDB.

CO - PO Mapping

Course Outcomes	Program Outcomes													
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	2	2	1	3	-	-	1	1	1	2	2	1	-
CO2	2	2	3	1	3	1	-	1	2	1	2	2	2	1
CO3	3	3	3	2	3	1	-	1	2	1	2	3	2	2
CO4	2	3	2	2	3	1	-	1	1	1	2	3	3	2
CO5	3	3	3	2	3	1	1	1	2	2	2	3	3	2
CO6	3	3	3	3	3	2	1	2	3	2	3	3	3	3

Practical No.	List of Practical's	Hrs.
1	**Title:** Installation and Configuration of Android Studio, Flutter SDK, VS Code, GitHub, and Emulator for Smart Food Delivery Mobile Application **Concepts:** Android Studio, Flutter SDK, Emulator Setup, VS Code, GitHub, Flutter Environment Configuration	02
2	**Title:** Creating First Flutter Mobile Application by Designing Smart Food Delivery Welcome Screen **Concepts:** Dart Basics, Flutter Widgets, Stateless Widget, Stateful Widget, UI Design, Material App	02
3	**Title:** Designing Responsive Login and Registration Screens for Smart Food Delivery Authentication System **Concepts:** Forms, TextField, Validation, Navigation, Responsive UI, User Authentication Interface	02
4	**Title:** Developing Smart Food Delivery Dashboard with Restaurant Cards and Bottom Navigation System **Concepts:** GridView, ListView, Drawer, Bottom Navigation, Routing, UI Components, Responsive Layout	02
5	**Title:** Building RESTful Flask APIs using Flask Framework for Smart Food Delivery Backend System **Concepts:** Flask API, REST API, CRUD Operations, JSON Handling, API Testing using Postman	02
6	**Title:** Integrating MongoDB Database with Flask APIs for Smart Food Delivery Order and Customer Management **Concepts:** MongoDB, Collections, Documents, Database Connectivity, PyMongo, Data Storage	02
7	**Title:** Consuming Restaurant and Delivery REST APIs in Flutter and Displaying Real-Time Food Data **Concepts:** API Integration, HTTP Requests, JSON Parsing, Future Builder, Async Programming	02
8	**Title:** Implementing JWT Authentication System for Secure Smart Food Delivery Mobile Application **Concepts:** JWT Authentication, Secure APIs, Login System, Token Validation, Authorization, Session Management	02
9	**Title:** Developing and Deploying Smart Food Delivery Backend APIs using Django REST Framework on Render/Railway **Concepts:** Django, Django REST Framework, Models, Serializers, CRUD APIs, Cloud Deployment, GitHub Integration, Environment Variables	02
10	**Mini Project (6 Hrs):** Developing and Deploying Full Stack Mobile Application Using Flutter, Flask/Django, and MongoDB **Concepts:** Full Stack Development, Frontend-Backend Integration, MongoDB Integration, Authentication, API Integration, Cloud Deployment, Live Hosting	06

References:

Text Books:

1. Miguel Grinberg, Flask Web Development: Developing Web Applications with Python, O'Reilly Media, Second Edition (or latest).
2. Marco L. Napoli, Beginning Flutter: A Hands-On Guide to App Development, Wrox (Wiley), Latest Edition.

Reference Books:

3. Dragan Gaic, Beginning Django: Delivering Military-Grade Web Development, Apress, Latest Edition.
4. Sanjeev Jaiswal and Ratan Kumar, Learning Django Web Development, Packt Publishing, Latest Edition.